SEPARATION OF INSOLUBLE PRODUCTS

FILTRATION

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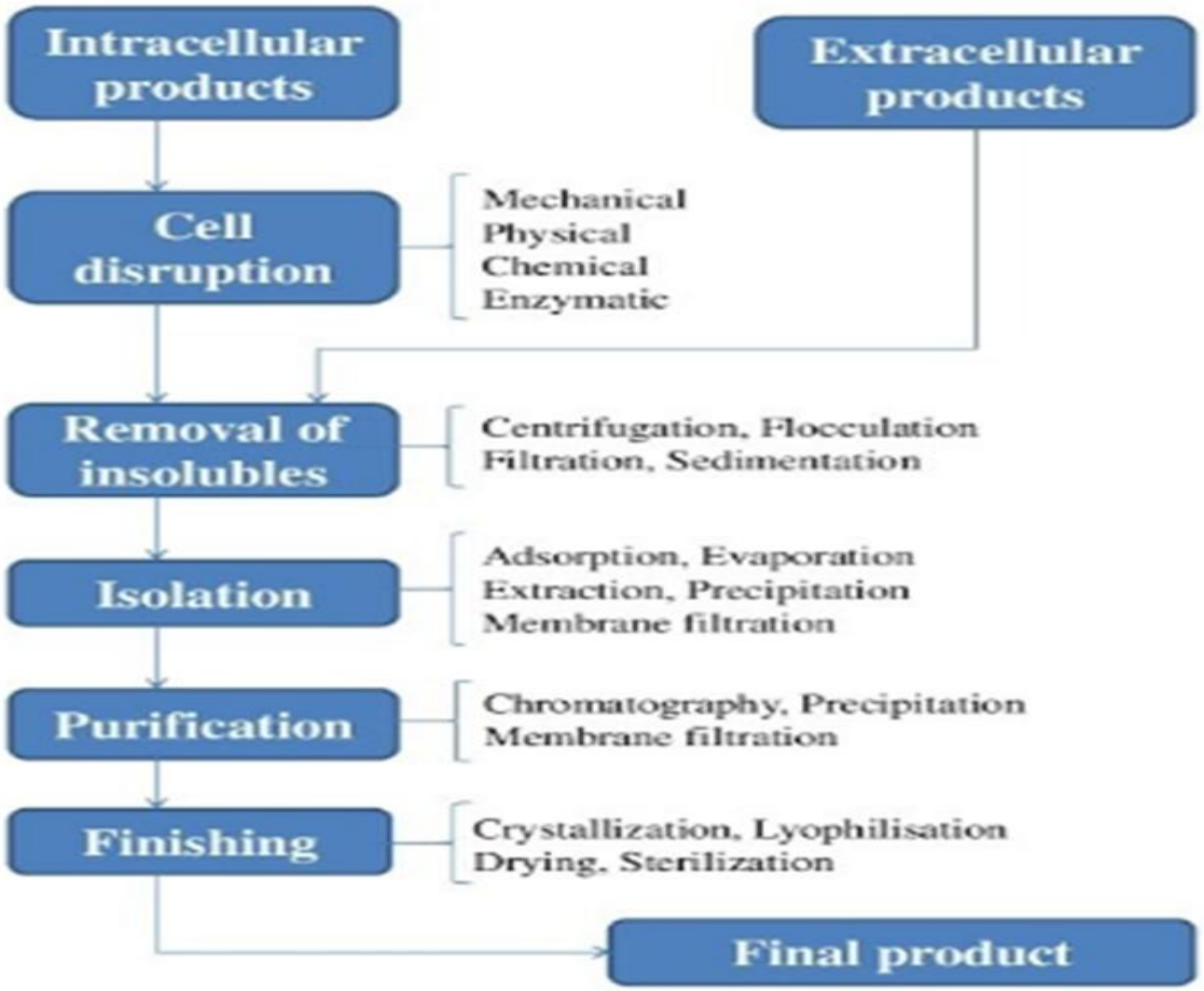
Downstream process



OVERVIEW

Overview

- **An insoluble substance is a substance (particle) that will not dissolve in a solvent even after mixing (The main insoluble considered are whole cells, cell debris, and particles such as inclusion bodies)**

1. Filtration
2. Sedimentation
3. Centrifugation
4. Coagulation
5. Flocculation

Filtration

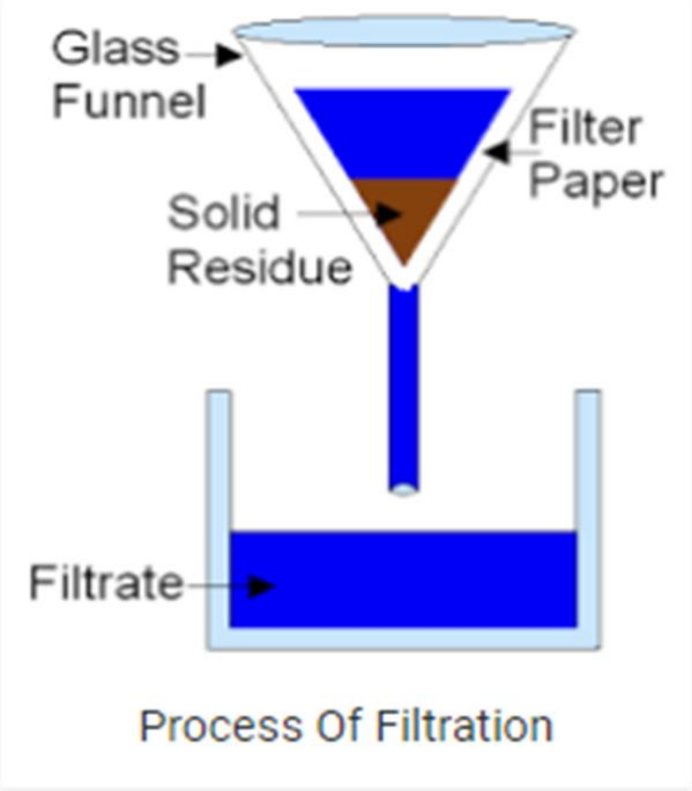
Inclusion bodies are aggregates of specific types of protein found in neurons, a number of tissue cells including red blood cells, bacteria, viruses, and plants

LEARNING OUTCOME

- Understanding, defining and knowing the application of the filtration process
- Explain the mechanism of the filtration process.
- Describe the type and function of the filter medium
- Differentiate between constant pressure cake and constant rate cake
- Investigating methods for increasing the efficiency of the filtration process
- Differentiate the mode of operation for the filtration process

Introduction

- **Filtration**, the **process** in which solid particles in a liquid or gaseous fluid are removed by the use of a porous medium that permits the fluid to pass through but retains the solid particles.
- The **porous medium which allows the liquid to go through** while retaining the solids is called the **filter**.
- The retained solid is called "**the residue**" or "**the cake**".
- The clarified liquid is called "**the effluent**" or the "**filtrate**"



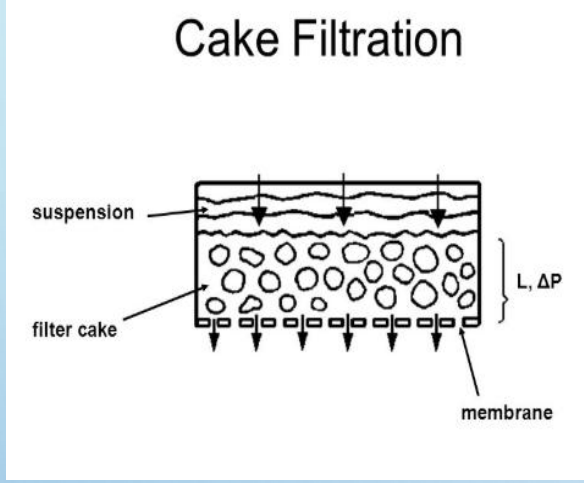
https://youtu.be/WSdGxQBnhZU

Applications of Filtration in Bioprocessing

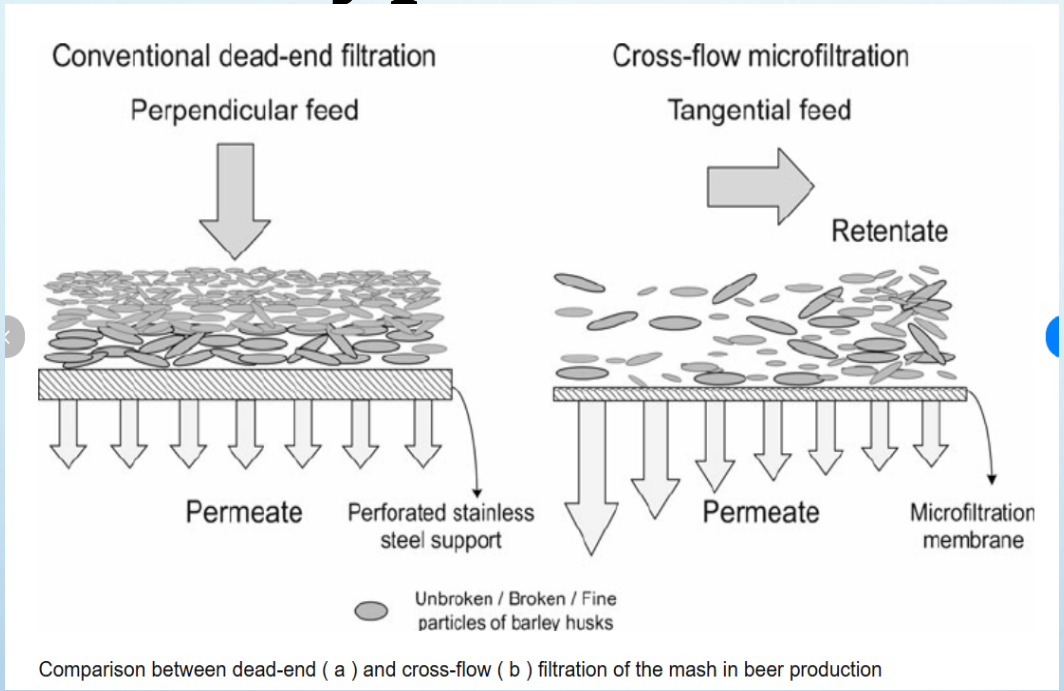
- Recovery of crystalline solids
- Recovery of cells from fermentation medium
- Clarification of liquids and gases
- Sterilization of liquids- https://www.youtube.com/watch?v=vcxy6FpfSuw

The term clarification is applied when the solid content in the feed does not exceed 1 wt %. In a clarification process, the filtrate is the primary product.

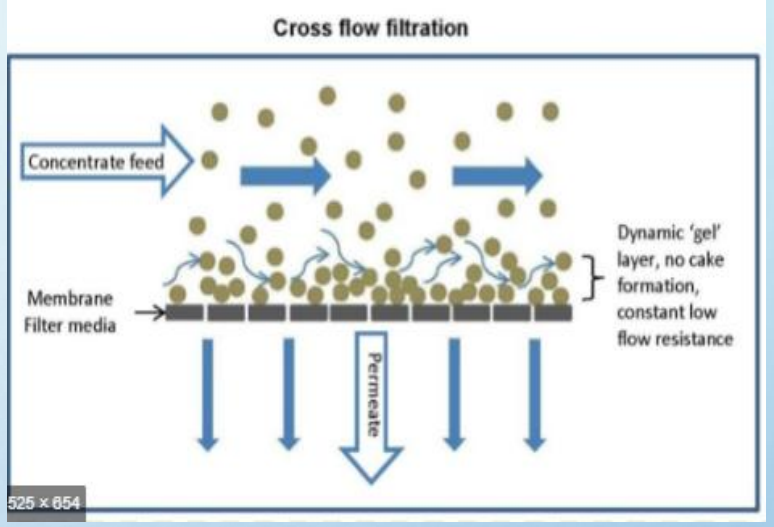
Types of Filtration



If recovery of solids from the high solid content slurry is desired, the process called cake filtration

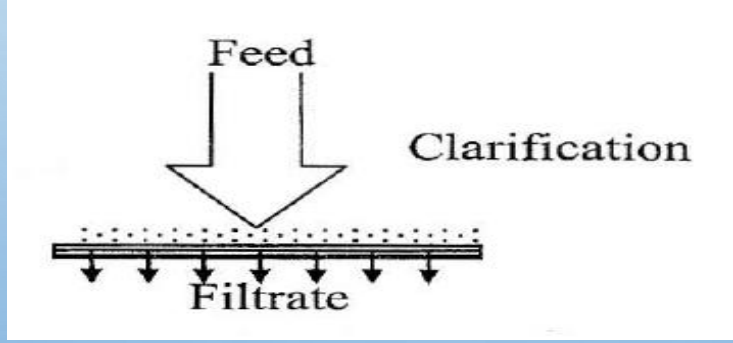


Comparison between dead-end (a) and cross-flow (b) filtration of the mash in beer production



Liquid flows parallel to the filtration medium

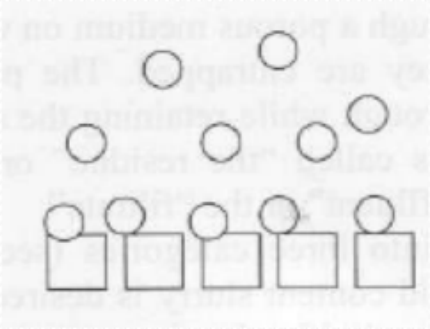
- Used for membrane filtration



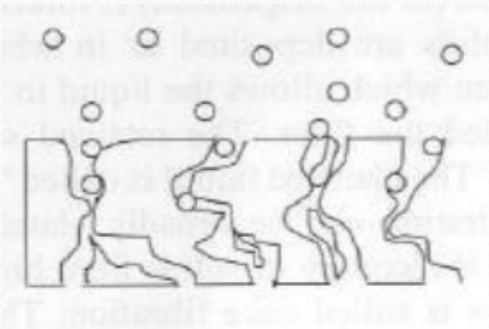
- **Solid content does not exceed 1 %**
- **Filtrate is the primary products**

Theory of filtration

- Two mechanism which solids are retained by a filter



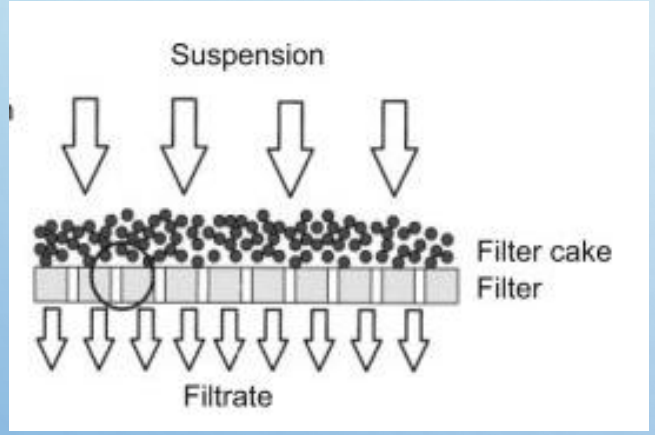
Surface filtration



Depth filtration

Mechanism of Filtration

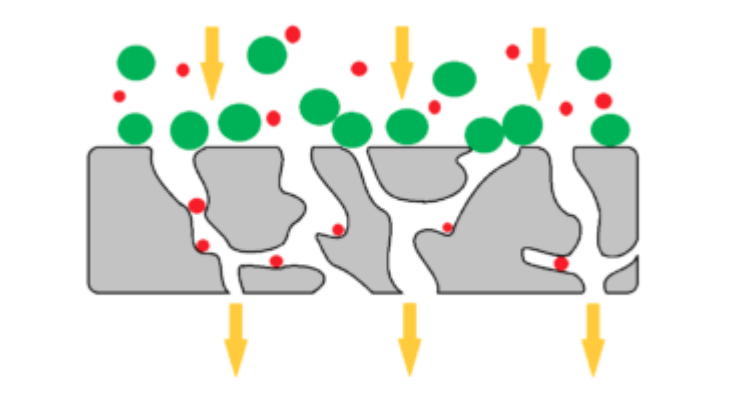
Surface Filtration



- Particles are retained by a screening action and held on the external surface of the filter
- Particles are not allowed to enter the filtration medium
- Pore size is precisely defined
- Only particles with diameter below pore size – permeate the filter
- Cake filtration & cross-flow filtration are based on surface filtration

Mechanism of Filtration

Depth Filtration



- Particles are allowed to penetrate pores and pore networks present in the filtration medium
- Do not have a precise pore size – not absolute
- Particles with wider range in size :

  Larger components – trapped on the filter surface

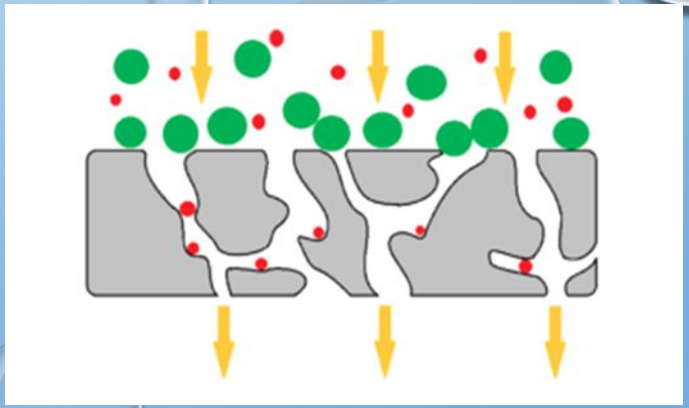
  Smaller particles – random entrapment & adsorption of matters occurs within the media structure
- Cheaper
- Particles retained within the filter by three mechanism:
  - direct interception
  - inertial impaction
  - diffusional interception.

Direct Interception

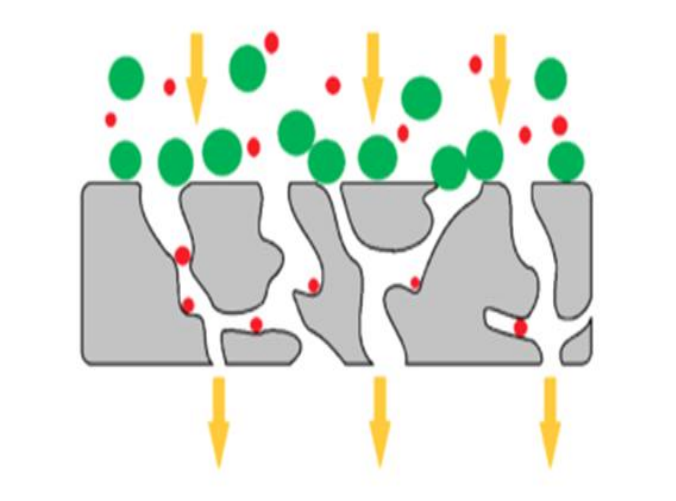
- **The particle enters the pores or pore networks within the filtration medium - gets trapped where the pore diameter becomes equal to the particle diameter**

  Pore diameter=particle diameter } Particles will be trapped

- **Particles that have diameters smaller than the pore diameter get trapped when pores are already constricted by collected particles.**



Inertial Impaction



- Pores in most filtration media are tortuous.
- Particles being carried by fluids possess momentum on their mass and velocity
- The flow of fluids through these pores is usually laminar in nature based on their small diameter.
- At tortuous sections - the fluid streamlines follows the curves while particles due to their inertia continue to move straight as a result hit the filter medium, lose their momentum, and are retained.
- This collection mechanism is referred to as inertial impaction.

Tortuous=full of twists and turns

Diffusional Interception

- Diffusional interception is more relevant to the filtration of the gas.
- The gas molecules, due to their random motion continually bombard suspended particles, particularly those smaller than 0.3 μm in diameter.
- The suspended particles deviate from their streamlines and impact the filter medium.
- Once this happens, the particles lose their momentum and are retained.
- Most clarification processes rely on depth filtration.

Filter Medium

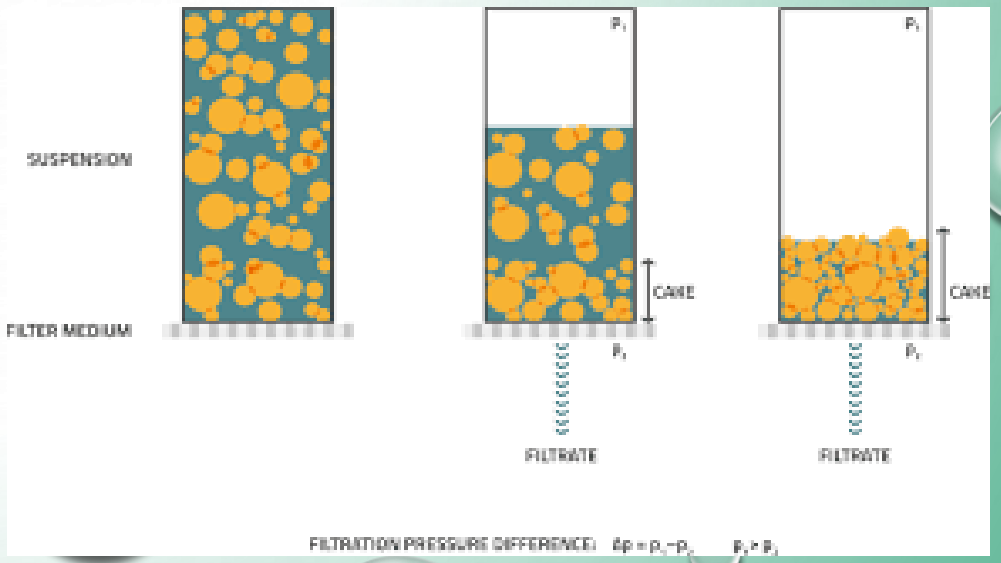
- The function of the filter medium is to act as barrier for particulate matter.

- At the beginning of a cake filtration process, the role of the filter medium is to act as a barrier. However, once the cake formation commences, the cake becomes the main particle-retaining barrier and the role of the filter medium is mainly as a support for the cake.

- The filter medium should have;

1. sufficient mechanical strength
2. resistant to the corrosive action of fluids being processed,
3. and offer low resistance to the flow of the filtrate.

- Commonly used filter media are:
  - Filter paper
  - Woven material
  - Non-woven fiber pads
  - Sintered and perforated glass
  - Sintered and perforated metal
  - Ceramics
  - Synthetic membranes



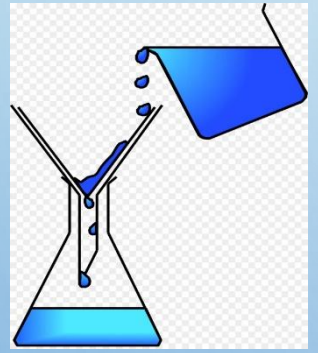
Driving Force

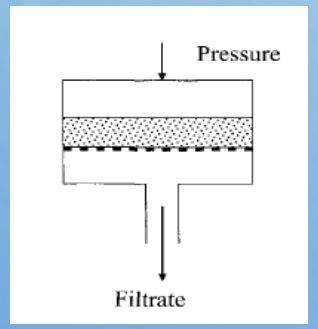
- Filtration is driven by applying a **pressure drop across the filter medium.**

- The **driving force** can be applied by:

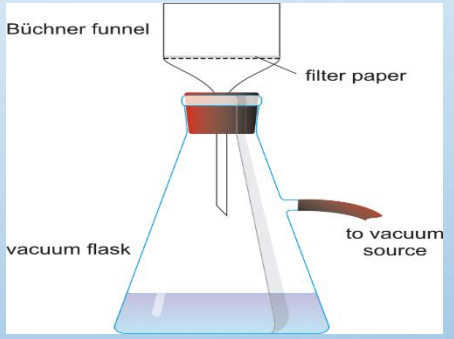
1. pressuring the feed side
2. by creating a vacuum in the filtrate side (vacuum filtration)

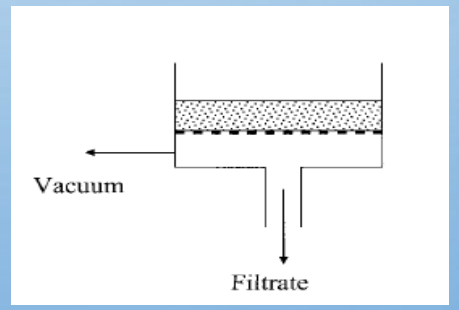
- **In the industry, both pressure and vacuum filtration are used.**





- **Pressure filtration** can be driven by pressurizing the feed using
  1. Compressed air pressurization
  2. Maintaining a hydrostatic liquid head on the feed side.
  3. The feed can also be pressurized with a suitable pump.





- **Vacuum filtration is commonly used in the laboratory.**
- **Vacuum can be generated using a water jet injector or a vacuum pump.**
- **Vacuum filtration is preferred from a safety point of view**
- **Substances that form foam cannot be filtered by vacuum filtration.**
- https://youtu.be/uxtXIvW4g5k

Constant Pressure Cake Filtration

- **Constant pressure filtration** refers to a filtration process where the **driving force(i.e. the pressure drop across the filter medium) is kept constant.**

- If we consider the filtration of a newtonian liquid, which is free from particles, the flow of liquid through the filter can be explained by **darcy's law.**

$$Q = \frac{kA\Delta P}{\mu l}$$

- Q = VOLUMETRIC FILTRATION RATE (m^3/s)
- k = DARCY'S LAW PERMEABILITY (m^2)
- A = AREA OF FILTER MEDIUM (m^2)
- ΔP = PRESSURE DROP ACROSS THE FILTER MEDIUM (Pa)
- μ = VISCOSITY ($kg/m\,s$)
- l = THICKNESS OF THE FILTER MEDIUM (m)

Pure ethanol is being filtered through a porous disc having a diameter of 2.5 cm and a thickness of 1 mm. When a pressure drop of 90 kPa was used to drive the filtration, the filtration rate obtained was 15 ml/min. If a porous disc made of the same material but having a diameter of 5 cm and a thickness of 3.5 mm were used for filtering ethanol, what filtration rate would you expect at 150 kPa pressure drop?

$$Q = \frac{kA\Delta P}{\mu l}$$

Q = volumetric filtration rate (m^3/s)
k = Darcy's law permeability (m^2)
A = Area of filter medium (m^2)
ΔP = Pressure drop across the filter medium (Pa)
μ = viscosity (kg/m s)
l = thickness of the filter medium (m)

Q1=15 ml/min
D1=2.5 cm
P=90 kpa
l=1 mm
u=?

D2=5 cm
l=3.5 mm
P=150 kpa
Q2=?

Q2=28.57 ml/min or 4.76×10^{-7} m3/s

Constant Rate Cake Filtration

Constant rate cake filtration

- Constant rate filtration refers to a filtration process where the filtration rate is kept constant by appropriately adjusting the presure drop during the process. When the filtration rate (Q) is constant, equation (6.3) can be written as:

$$Q = \frac{A\Delta P(t)}{\mu(R_m + \frac{\alpha C_a Q t}{A}} \qquad (6.4)$$

- Where AP(t)=instantaneous pressure drop (Pa)
- Rearrangging equation (6.4) we get:

$$\Delta P(t) = \frac{\mu\alpha C_a Q^2 t}{A^2} + \frac{\mu Q R_m}{A} \qquad (6.5)$$

- When the medium resistance is negligible compared to the cake resistance, we can write equation (6.5) as:

$$\Delta P(t) = \frac{\mu\alpha C_a Q^2 t}{A^2}$$

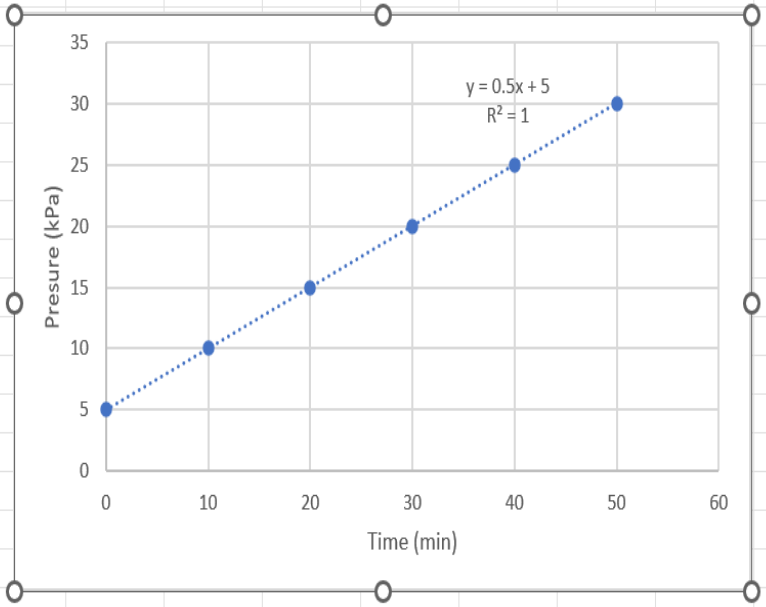
- Question2

Time-pressure drop data for a constant rate cake filtration process is given in the table below. If we are to carry out the filtration at twice the filtration rate, predict the pressure drop required after 1 hour of filtration. Assume that the cake is incommpressible.

Time (min)	0	10	20	30	40	50
ΔP (kPa)	5	10	15	20	25	30

Ans: ΔP=130 kPa

time	pressure
0	5
10	10
20	15
30	20
40	25
50	30



$$\Delta P(t) = \frac{\mu \alpha C_a Q^2 t}{A^2} + \frac{\mu Q R_m}{A}$$

y=0.5t + 5

-next filtration will be carried out at twice the filtration rate =2 x Q

-slope = 2^2=4 times the equation

-intercept= 2 times the equation

y=2t + 10 =2 (60 min) + 10 = 130 kPa

Improvement of Filtration Efficiency

- Efficiency of cake filtration depends on achieving high cake accumulation on the filter medium
- However, the filtration rate declines with cake accumulation due to an increase in cake resistance

How to overcome ???

- Feed pre-treatment
- Filter aid

Improvement of Filtration Efficiency

Feed pre-treatment

- The feed can be treated by physical method (ex: heating-lower the viscosity) or by the addition of chemicals (coagulants/flocculants) to obtain porous cake with low specific cake resistance.

- Drawback – thermolabile substance cannot be heated, addition of chemical not possible in some applications

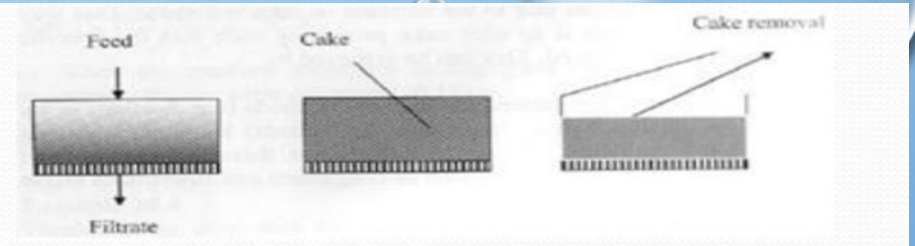


Fig 10.6 Batch cake filtration

Improvement of Filtration Efficiency

Filter-aids

- Filter aids are substances that are mixed with feed for creating very porous cakes
- Filter aids are granular solids that are used in combination with filtration to enhance the rate of filtration
- This increases the filtration rate very significantly
- The filter aids can later be removed from the dried and powdered cake by suitable separation technique- sieving
- Rarely used when the cake is the product of interest

Filter aids are typically fine, light-colored powders that are added to a liquid to improve the porosity of the filter cake. The most common types are Diatomaceous Earth (made from fossilized algae) and Perlite (volcanic glass).

Factors influencing filtration

Properties of solid	Properties of liquids	Properties of solid in slurry	Objectives	Temperature
• Particle shape, size and charge • Density • Particle size distribution • Rigidity or compressibility of solid under pressure • Tendency of particle to flocculate or adhere together	• Density • Viscosity • Corrosiveness	• Rate of formation of filter cake especially in early stage of filtration	• Whether the solid or liquid or both are to be collected	• Temperature of suspension

Mode of Operation

- **Filtration can be carried out in different ways;**

1. **Filters can be operated as a batch, where the cake is removed after a run**
2. **Continuous, where the cake is removed continuously.**
3. **Slurry thickening by delayed cake filtration**
4. **Slurry thickening by cross-flow filtration**
5. **Cake washing**

Cake accumulation and removal in batch mode.

- The **common mode** for **small-scale cake filtration**.
- A **batch of feed is pumped into the filter unit** and **filtration is carried out either** at a **constant rate** or at **constant pressure.**
- The **process is terminated** when the **filtration rate gets unacceptably low**, or when **the pressure required gets too high**, or when the **filtration device is filled with the filter cake**.
- The **cake is then removed from the device by scraping it off the filter medium.** Often this requires the dismantling of the filtration unit.
- The **filter medium is usually then cleaned and made ready for the next batch.**
- Examples of filtration devices operated in this mode include:
- Funnel filter
- Filter press
- Leaf pressure filter
- Vacuum leaf filter

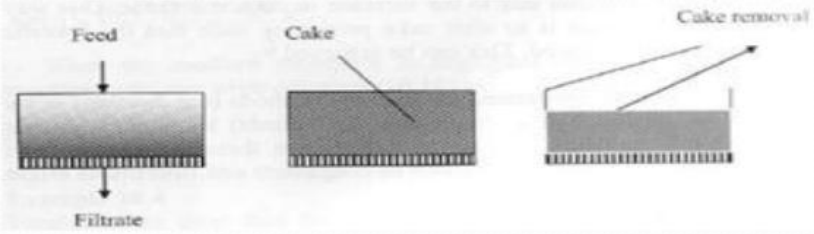
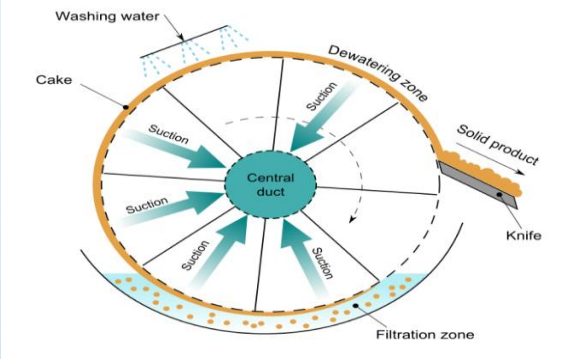


Fig 10.6 Batch cake filtration

Continuous cake accumulation and removal.

- A **batch process may not be suitable** for **large-scale cake filtration.**
- **Continuous filters** which **allow simultaneous cake accumulation and removal** are usually used for **large-scale processes**.
- Examples of such:
  - **Rotary drum filter-**How It Works (Step-by-Step)



Filtration Zone

- The lower portion of the rotating drum is submerged in the slurry.
- The outer surface of the drum is covered with a filter cloth.
- Inside the drum, there are several compartments connected to a central duct through pipes.
- As each compartment passes through the slurry, a vacuum is applied via the central duct (connected to a vacuum pump). The liquid (filtrate) is sucked through the filter cloth into the drum and collected via the central duct. The solid particles remain on the outside of the cloth, forming a filter cake.

Washing Zone

- The drum continues to rotate. Once the cake leaves the slurry, washing water is sprayed on it. This removes impurities or soluble substances from the cake using the same vacuum suction pulling the wash liquid through the cake.

Dewatering Zone

- Further rotation exposes the cake to air while the vacuum continues to pull moisture out. This dries (dewaters) the cake before discharge.

Cake Discharge Zone

- At the top of the rotation, a knife blade or scraper removes the dried solid cake from the drum surface. The clean section of the drum then re-enters the slurry, and the cycle repeats continuously.

Slurry thickening by delayed cake filtration

- The main objective here is not to separate solids completely, but to thicken the slurry , that means increasing the solid concentration in the liquid without allowing a thick filter cake to form on the filter medium.

- In normal filtration, solids accumulate on the filter medium and form a cake that grows thicker over time.

- In delayed cake filtration, we try to avoid that thick cake build-up , we only allow a very thin layer or continuously remove it.

- **Slurry thickening** can be **achieved by controlling the thickness of the cake layer** thereby keeping the **particulate matter in a suspended form on the feed side.**

- This can be achieved by **incorporating mechanical devices such as moving blades,** which **continuously scrape off the cake from the filter surface.**

- With the **moving blade arrangement**, the **thickness of the cake is limited by the clearance between the filter medium and the blade.**

- This type of **filtration can be carried out until the solid content on the feed side reaches a critical level beyond which the slurry does not flow.**

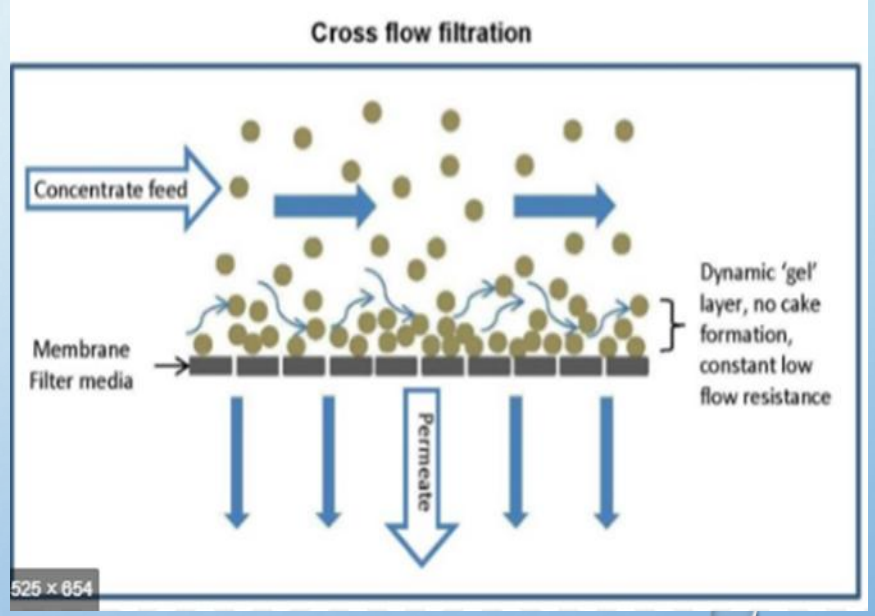
Slurry thickening by cross-flow filtration

- An alternative way by which the **build-up of cake on the filter can be reduced** is by using a **cross-flow mode of operation.**

- This is achieved **by maintaining a very high velocity of feed flow parallel to the surface of the filter medium.**

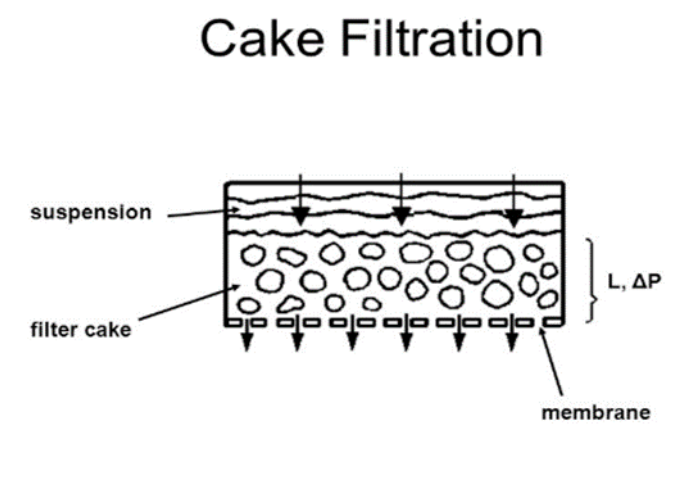
- Typical cross-flow rates may be 10 - 20 times the filtration rate.

- However, **cross-flow filtration cannot be used for obtaining very thick slurry** since the **energy required for maintaining the cross-flow velocity becomes prohibitively high.**



Cake washing

- After its formation, the cake may contain a significant amount of entrapped liquid.
- When the **liquid is the product of interest,** this entrapment represents a loss of yield.
- When the **cake is itself the product,** the entrapped liquid represents the presence of impurity.
- The **entrapped liquid can be removed by cake washing.**
- The **washing liquid** should not be a **"new impurity".**
- Its presence should either be acceptable in **the final product** (i.e. either the cake or the filtrate), or it should be easily **"removable".**



- The choice of washing liquid depends on the **nature of the product, type of impurities,** and **whether the cake or the filtrate** is the desired product.

Filter Equipment

- *Filter Press*

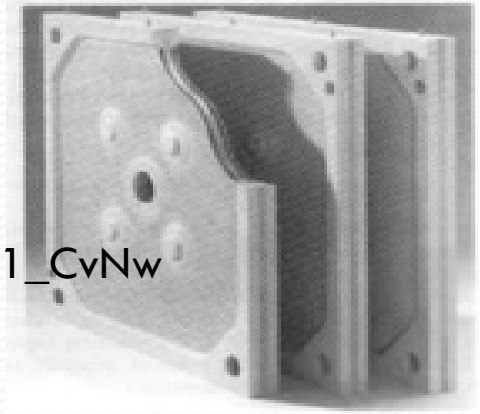
A filter press consists of a series of horizontally arranged vertical filter elements, each consisting of a frame within which the cake can be accumulated sadwiched between filter medium on either side. Each filter medium is supported on a plate which has grooves to allow easy collection of the filtrate.

A plate and frame filter press is mainly used for cake filtration and cake washing.

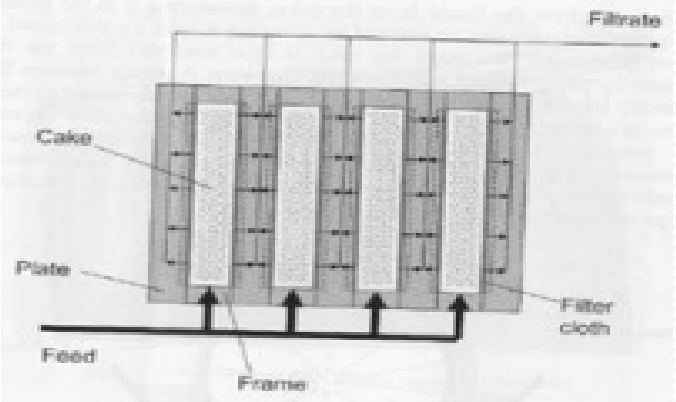
Advantages-compact design and high throughput

Disadvantages-High labor cost, high down times, capacity limitation and batch-wise operation.

Filter press plates



https://www.youtube.com/watch?v=M4wBd1_CvNw



- *Rotary Drum Vacuum Filter*

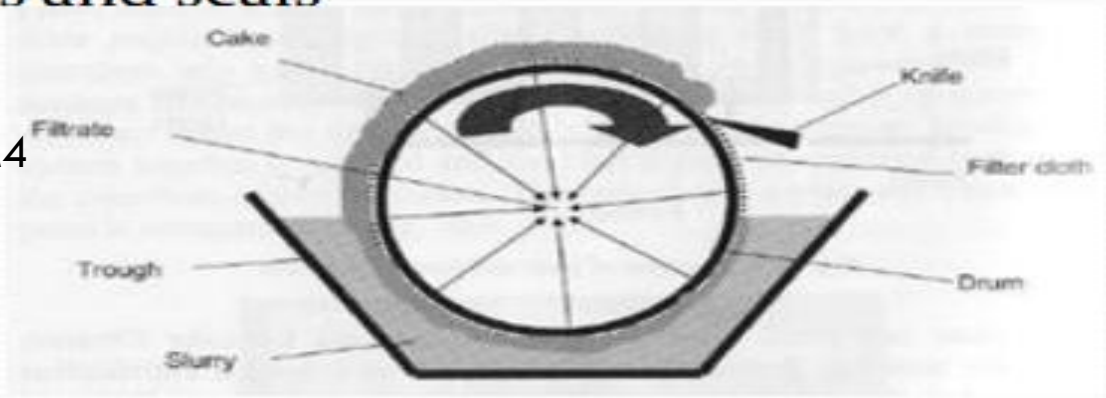
A rotary drum vacuum filter is a continuous filtration device in which the solids are separated by a porous filter cloth or similar filtration media wrapped around a drum-like structure with a perforated curved surface.

Used for cake filtration, cake washing and dewatering

Advantages-continuous operation and cake drying

Disadvantages-low driving force and complicated design with many moving parts and seals

https://www.youtube.com/watch?v=m2d-Ov9RIM4



The working principle of a rotary drum filter

- *Pressure leaf filter*

A pressure leaf filter consist of a number of rectangular basic filtration units (also called leaves) connected together in parallel by means of flexible hose or a rigid tube manifold.

Used for cake filtration and clarification.

Advantages-simplicity of design and flexibility of use.

Disadvantages-high labor cost and high equipment foot-print.

Pressure leaf filter

D1=2.5 cm= 0.025 m, l1=1mm=0.001 m, $\Delta P1 = 90\ kPa, Q1 = 15\frac{ml}{min}$

D2=5 cm, l2= 3.5 mm= 0.0035 m , $\Delta P2 = 150\ kPa, Q2 = ?$

Use darcy law equation..
μ and k constant for both situation because same material and fluid used

Q1 divide Q2

A=$\frac{\pi}{4}d^2$
Q2=28.57 ml/min